

BSc Graduation Project Report

AI-Based Academic Decision Support System

Predicting Student Performance, Dropout Risk, and
Academic Support Needs

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Project Flow at a Glance

Student Data grades, absences	Preprocess clean + encode	ML Model GB/AdaBoost	AI Agent explain + advise	Advisor support plan
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Examiner Comment Coverage

Requested improvement	Where addressed
Clearer literature review and previous studies comparison	Chapter 3, Sections 3.2.1 and 3.2.2
Detailed dataset explanation	Chapter 2, Section 2.3.1
Implementation steps and evaluation process	Chapter 6, Section 6.2.1

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Main Deliverable Coverage Summary

Required item	Where covered	Status
Problem and objectives	Chapter 1	Covered
Datasets and requirements	Chapter 2 + 2.3.1	Expanded
Literature review and research gap	Chapter 3 + 3.2.1-3.2.2	Expanded
Implementation and evaluation	Chapter 4 + Chapter 6	Expanded

ABSTRACT

Students and academic institutions generate large amounts of educational data, but this data is often not fully used to support academic decision-making. Many students face academic risks such as low performance, delayed academic progress, or dropout. This project proposes an AI-based Academic Decision Support System that applies machine learning to predict student academic risk and converts model outputs into understandable recommendations for students and academic advisors.

The project uses two required datasets: Dataset A, the UCI Student Performance Dataset, as the main training dataset, and Dataset B, the Student Dropout and Academic Success Dataset, as the validation, adaptation, and comparison dataset. The proposed methodology follows the data science lifecycle: problem understanding, data collection and cleaning, exploratory data analysis, feature engineering, model building, evaluation, and communication of findings. The system also includes AI-agent logic, a simple interface, and a database or function layer for integration.

The methodology starts with baseline models and then compares improved models such as Random Forest, Gradient Boosting, AdaBoost, and optional LightGBM if allowed. Explainability will be handled through feature importance, with SHAP and probability-calibration measures considered as optional enhancements if time allows. The expected result is a prototype that supports early identification of at-risk students and provides practical academic recommendations while keeping final decisions under human academic supervision.

Keywords: academic decision support, student performance, dropout prediction, machine learning, AI agent, explainable AI

CHAPTER 1: INTRODUCTION

1.1 Purpose of the Project

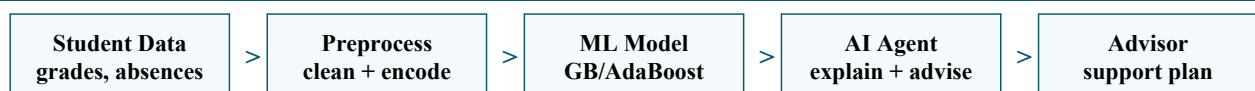
The purpose of this project is to develop an AI-based system that supports academic decision-making by predicting student performance and academic risk. The project applies machine learning to educational datasets and converts model predictions into clear explanations and recommendations through an AI-agent layer.

1.2 Project Background and Motivation

Educational institutions collect student information such as grades, attendance, demographic attributes, academic progress, and social indicators. However, the value of this data is limited if it is not analyzed systematically. Academic advisors need early signals that help them identify students who may need support. Machine learning can detect patterns in student data and help predict whether a student is likely to succeed, remain in progress, or become at risk.

The motivation for this project is to transform raw student data into decision support. Instead of providing only a prediction, the proposed system will explain the prediction and recommend academic actions such as advising, monitoring attendance, or providing additional learning support.

Figure 1.1: Academic risk prediction flow



This figure summarizes the project idea from student data to advisor action.

1.3 Problem Statement

Students and institutions have academic data but often struggle to make effective academic decisions from it. The main problem is how to use student data to predict academic outcomes and provide practical recommendations that help students and advisors take early action. Many academic risks are discovered late, after the student has already failed a course, delayed progress, or reached a weak academic status. Therefore, there is a need for an intelligent support system that can analyze student data, identify risk indicators, and present recommendations in a clear form.

1.4 Project Scope

- Use both mandatory datasets: UCI Student Performance Dataset for training and Student Dropout and Academic Success Dataset for validation/adaptation.
- Build machine-learning models for academic outcome and academic-risk prediction.

- Compare baseline models and advanced ensemble models.
- Handle different dataset structures through target mapping and feature alignment.
- Add AI-agent logic for explanation and recommendation.
- Integrate the model with a function/database and a simple user interface.
- Evaluate performance and compare results across both datasets.
- Document limitations, ethical considerations, and future improvements.

1.5 Objectives

- Understand the academic decision-making problem and define measurable success criteria.
- Collect, clean, and analyze the required datasets.
- Build an initial model in GP1 and improve it in GP2.
- Evaluate models using accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC, and probability reliability where applicable.
- Use explainable AI or feature importance to show why a student is classified as at risk.
- Design an AI agent that produces explanations and recommendations.
- Build a prototype interface and integrate it with the model and storage layer.
- Compare the two datasets and discuss challenges, adjustments, and performance differences.

1.6 Expected Outcomes

- A cleaned and preprocessed version of both datasets.
- EDA charts and statistical summaries.
- Baseline and improved machine-learning models.
- A dataset comparison section that explains differences and adjustments.
- A simple AI agent that returns risk level, explanation, and recommendation.
- A simple interface for demonstration, preferably Streamlit.
- A final report, presentation, source code, and demo instructions.

CHAPTER 2: SYSTEM ANALYSIS AND REQUIREMENTS

2.1 Data Science Lifecycle Alignment

The project follows a structured data science lifecycle. This is important because the project is not only a programming task; it is a decision-support process that begins with a real problem and ends with useful recommendations [11].

Figure 2.1: Data science lifecycle adapted to the project



The project aligns with the lifecycle from problem understanding to decision support.

Lifecycle Stage	How It Is Applied in This Project
Problem understanding	Define the academic decision problem and success criteria, such as detecting at-risk students early.
Data collection and cleaning	Download Dataset A and Dataset B, handle missing values, duplicates, and inconsistent formats.
Exploratory data analysis	Study distributions, class balance, correlations, and relationships between features and outcomes.
Feature engineering	Create academic risk categories, progress indicators, and aligned features across datasets.
Model building and selection	Train baseline and advanced classification models and compare them.
Evaluation	Use confusion matrix, accuracy, precision, recall, F1-score, ROC-AUC, and calibration measures if implemented.
Communication and visualization	Provide charts, model explanations, risk levels, and academic recommendations through the interface.

2.2 Dataset A and 2.3 Dataset B Overview

Dataset A is used for initial student-performance and academic-risk prediction. Dataset B is used for dropout, enrollment, and graduation validation. Since the two datasets have different structures, the project aligns them through common academic-risk categories rather than claiming a direct transfer.

Item	Dataset A - Student Performance	Dataset B - Dropout and Academic Success
Source / size	UCI Machine Learning Repository; 649 student records in the commonly used Portuguese subset.	UCI Machine Learning Repository; 4,424 records and about 36 attributes.
Type / features	Structured academic, demographic, social, and school-related data: grades, absences, failures, study time, family support.	Structured demographic, enrollment, financial, and semester-progress data: approved units, tuition status, scholarship, admission grade.
Target / justification	Final grade or derived risk category; suitable for training initial risk prediction models.	Dropout, Enrolled, Graduate; suitable for GP2 validation because it directly supports early-warning outcomes.
Preprocessing	Categorical encoding, grade-to-risk mapping, selected scaling, and feature selection.	Target mapping, class-balance check, categorical encoding, scaling, and possible imbalance treatment.

2.3.1 Detailed Dataset Explanation

The dataset selection is justified because Dataset A explains student performance patterns, while Dataset B evaluates continuation and dropout outcomes. Together, they support a decision-support system that predicts risk, explains important factors, and recommends academic support actions.

Dataset A: final grade -> risk	Common Mapping: At Risk In Progress Successful	Dataset B: dropout / enrolled / graduate
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2.4 Functional Requirements

Code	Requirement	Code	Requirement
FR1	Load Dataset A and Dataset B from CSV files.	FR6	Generate explanation using feature importance or SHAP where possible.
FR2	Clean and preprocess numerical and categorical features.	FR7	Provide recommendation through the AI-agent layer.
FR3	Train baseline and improved models.	FR8	Store or export prediction results for review.
FR4	Evaluate performance using classification metrics.	FR9	Allow users to enter/select student information and view results.
FR5	Map model output into academic risk levels.		

2.5 Non-Functional Requirements

Category	Requirement
Usability	The interface should be simple and understandable for students, advisors, and examiners.
Performance	Prediction should be produced within a few seconds for demo-scale data.
Reliability / Maintainability	The same preprocessing pipeline should be used during training and prediction; code should be modular.
Security, privacy, and ethics	If real student data is used later, anonymization, access control, and human review should be applied.

CHAPTER 3: LITERATURE REVIEW AND RESEARCH-BASED ENHANCEMENTS

3.1 Related Work Summary

Previous studies show that academic-risk and dropout problems can be supported using machine-learning methods. The reviewed work guides model comparison, imbalance handling, probability-based risk levels, and explainable recommendations for advisors.

The literature review is organized around three themes: student performance prediction, dropout and academic-success prediction, and explainable decision support. This organization makes the review easier to connect to the selected datasets, models, and advisor-facing output.

3.2 Research Gap Addressed by This Project

Many academic-performance projects stop after predicting a grade, dropout class, or risk score. However, advisors need more than a label. They need a clear explanation and a practical support action that can be discussed with the student.

3.2.1 Clarified Research Gap

This project addresses the gap by combining prediction, explanation, and advisor-friendly recommendation in one decision-support workflow. The model output is not intended to replace academic judgment. Instead, it provides early signals and support suggestions that advisors can review before making decisions.

Advisor-friendly interpretation

- The system should identify whether a student is Low, Medium, or High Risk.
- The system should explain the main factors behind the prediction, such as grades, absences, failures, or progress indicators.
- The system should recommend a practical action, such as monitoring, academic advising, attendance follow-up, or an intervention plan.

Literature review organization

Theme	Purpose in this project
Student performance prediction	Supports Dataset A and grade-to-risk mapping.
Dropout and academic-success prediction	Supports Dataset B validation and comparison.
Explainable decision support	Connects model outputs with advisor-friendly recommendations.

3.2.2 Organized Literature Review and Previous Studies

The following table summarizes the most relevant previous studies. It highlights the key findings and explains how each study supports the design choices in this project.

Study	Focus / Methods	Key Findings and Relevance
Cortez and Silva / UCI Student Performance	Student achievement prediction using academic, school, and family features.	Grades, failures, study time, and absences support Dataset A and grade-to-risk mapping.
Vafeiadis et al. (2015)	Compared churn-prediction models including SVM, decision trees, logistic regression, and boosting.	Boosting improved several classifiers, supporting Gradient Boosting/AdaBoost comparison.
Vaduva et al. (2024)	Used Random Forest, LightGBM, SMOTETomek, calibration, SHAP, F1-score, and ROC-AUC.	Supports imbalance handling, explainability, and probability-based risk levels.
Dropout and academic success studies	Predict dropout, enrollment, and graduation using enrollment, financial, and semester-progress features.	Supports Dataset B for GP2 validation and comparison.

Key comparison between studies

Finding from review	Action in GP2
Baseline models are simple and interpretable.	Train Logistic Regression and Decision Tree for comparison.
Ensemble models often improve tabular-data performance.	Compare Random Forest, Gradient Boosting, and AdaBoost.
Dropout datasets may be imbalanced.	Check class balance and apply class_weight, SMOTE, or SMOTETomek if needed.
Academic decisions require transparency.	Use feature importance or SHAP and convert output into advisor-friendly language.

3.3 Enhancements Added After Review

Enhancement	Why It Was Added	How It Will Be Used
Model comparison	Avoid dependence on a single model.	Compare baseline and ensemble models.
Class-imbalance handling	Dropout datasets often have imbalanced classes.	Use class_weight, SMOTE, or SMOTETomek if needed.
Explainable AI	Academic decisions require transparency.	Use feature importance or SHAP to identify top risk factors.
Ethical decision support	Predictions should not replace human judgment.	Add human-review warning and advisor supervision.

CHAPTER 4: METHODOLOGY AND DESIGN CONSIDERATIONS

4.1 Proposed Methodology

The proposed methodology follows the data science lifecycle. It starts with understanding the academic-risk problem, then prepares the datasets, trains the models, evaluates results, and converts predictions into explanations and recommendations.

- Download Dataset A and Dataset B in CSV format and inspect columns, missing values, target labels, and class distributions.
- Clean the datasets by handling missing values, duplicated rows, outliers, and inconsistent formats.
- Encode categorical variables and scale numerical variables when required by models such as SVM and Logistic Regression.
- Create common target mappings such as At Risk, In Progress, and Successful to support comparison between datasets.
- Train baseline and improved models, evaluate them, and select the best model based on the project goal.
- Use explainability methods and AI-agent rules to convert the prediction into a clear explanation and recommendation.

Methodology phases and GP2 output

Phase	Activity	Expected GP2 Output
Data preparation	Load, clean, encode, and align Dataset A and Dataset B.	Clean training and validation datasets.
Modeling	Train baseline and improved machine-learning models.	Model comparison results.
Evaluation	Use accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC where applicable.	Clear performance evidence.
Decision support	Convert prediction into risk level, explanation, and recommendation.	Advisor-friendly support output.

This workflow prevents the system from becoming a black-box prediction tool and links the technical model to academic advising needs.

Planned GP2 artifacts

Artifact	Purpose
Trained model files	Demonstrate that the prediction pipeline was implemented.
Evaluation charts	Show performance using accuracy, precision, recall, F1-score, and confusion matrix.
Streamlit interface screenshots	Show how advisors can view risk level, explanation, and recommendation.

4.2 Algorithm Selection and Justification

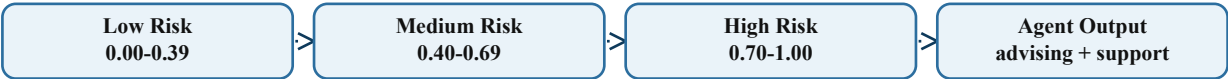
Algorithm	Role in Project	Reason for Selection
Logistic Regression	Baseline model	Simple, interpretable, and useful for comparison.
Decision Tree	Interpretable baseline	Produces simple rules and is easy to explain.
XGBoost / Gradient Boosting	Main selected model	Boosting can improve classification by focusing on difficult cases.
Random Forest	Comparison improved model	Handles nonlinear relationships and provides feature importance.
SVM	Alternative model	Useful for classification with scaled features and nonlinear boundaries.
LightGBM	Optional advanced model	Efficient gradient-boosting model if installation and course rules allow it.

4.3 Feature Engineering Plan

- Convert final grade in Dataset A into academic risk categories.
- Use previous grades, semester grades, approved units, and absences as academic progress indicators where available.
- Create engagement indicators from attendance, failures, or approved curricular units.
- Use correlation, feature importance, or SHAP to remove weak or redundant variables.
- Align common concepts across datasets even when exact column names are different.
- Avoid using sensitive features as the only reason for risk classification.

4.4 Risk Level Logic

Predicted Probability of Risk	Risk Level	Recommended Agent Output
0.00 - 0.39	Low Risk	Student is likely to continue successfully. Continue monitoring normally.
0.40 - 0.69	Medium Risk	Student may need support. Recommend monitoring grades and academic engagement.
0.70 - 1.00	High Risk	Student is likely at risk. Recommend academic advising and an intervention plan.



Risk levels are translated into advisor-friendly actions.

4.5 Alternative Solutions Considered

Alternative	Advantages	Limitations	Decision
Manual advisor review only	Human judgment and context-aware decisions.	Slow and may miss early risk patterns.	Not enough as the main solution.
Rule-based system only	Simple and easy to explain.	Rules may not capture complex patterns.	Use rules only in the AI-agent layer.
Machine-learning prediction only	Can detect patterns from data.	Prediction without explanation is less useful.	Combine ML with explanations and recommendations.
Deep learning model	Can model complex relationships.	May be harder to explain and unnecessary for tabular data.	Keep as future work, not GP1 main model.

CHAPTER 5: SYSTEM DESIGN

5.1 System Architecture

The proposed system is organized into five main layers: data layer, preprocessing layer, machine-learning layer, AI-agent layer, and interface/database layer.

Layer	Main Function
Data layer	Stores Dataset A, Dataset B, and optional prediction records.
Preprocessing layer	Applies cleaning, encoding, scaling, and feature engineering.
Machine-learning layer	Trains models and produces class predictions and probabilities.
Explainability layer	Produces feature importance or SHAP-based explanations.
AI-agent layer	Converts prediction and explanation into risk level and recommendation.
Interface/database layer	Allows user input, displays results, and stores outputs.

Figure 5.1: System architecture layers

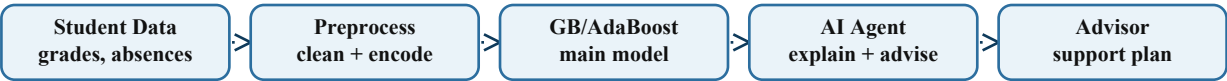


Layered architecture keeps the system modular and easier to maintain.

5.2 Program Flow

Student Data Input -> Preprocessing Pipeline -> ML Model Prediction -> Risk Probability -> Explainability Module -> AI Agent Recommendation -> Interface Output -> Database Record

Figure 5.2: Program flow diagram



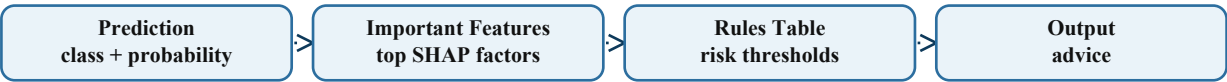
This diagram summarizes the operational path of the system.

5.3 AI Agent Logic

The AI agent is a rule-based decision-support layer. It receives the model prediction, predicted probability, and top contributing features. It then generates a readable explanation and recommendation. The agent will not make final academic decisions; it only supports students and advisors.

Input to Agent	Processing Logic	Output
Prediction class	Map prediction to success, in progress, or risk.	Academic status.
Predicted probability	Apply probability thresholds.	Low, Medium, or High risk.
Important features	Identify strongest positive risk factors.	Explanation in simple language.
Rules table	Match risk level to academic action.	Recommendation.

Figure 5.3: AI-agent decision logic



The AI-agent layer converts model outputs into language advisors can use.

5.4 Example Agent Output

The student is classified as High Risk with a probability of 78%. The most important factors are low previous academic performance, high absences, and limited academic progress. Recommended action: schedule an academic advising session, create a study-support plan, and monitor the student again after two weeks.

5.5 Suggested Interface Design

Page / Section	Purpose
Home page	Explain project goal and dataset sources.
Student input page	Enter or upload student features.
Prediction page	Display predicted class, probability, risk level, and confidence.
Recommendation page	Show AI-agent explanation and recommended action.
Dataset comparison page	Show evaluation metrics and comparison charts for both datasets.

CHAPTER 6: IMPLEMENTATION AND VALIDATION PLAN

6.1 Tools and Technologies

Category	Tools
Programming language	Python
Data analysis	pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine learning	scikit-learn; optional LightGBM if allowed
Class balancing	class_weight, SMOTE, or SMOTETomek
Explainability	Feature importance or SHAP
Model saving	Joblib or Pickle
Database	SQLite
Interface	Streamlit
IDE / Platform	Jupyter Notebook, Google Colab, or VS Code

6.2 Evaluation Metrics

Metric	Purpose
Accuracy	Measures overall correct predictions.
Precision	Shows how many predicted at-risk students are actually at risk.
Recall	Shows how many actual at-risk students were detected. This is very important for early intervention.
F1-score	Balances precision and recall.
Confusion matrix	Shows true positives, false positives, true negatives, and false negatives.
ROC-AUC	Measures class separation ability across thresholds.
Brier Score	Optional metric for probability reliability and calibration.

Figure 6.1: Evaluation Metrics Map



Evaluation focuses on recall because missing an at-risk student is more serious.

Figure 6.2: Confusion Matrix Interpretation

6.2.1 GP2 Implementation and Evaluation Details

This subsection explains how the solution will be implemented and evaluated in GP2. It expands the methodology, implementation steps, and evaluation process requested by the examiner.

Implementation Methodology

- Define the academic-risk task and advisor support objective.
- Load Dataset A and Dataset B, inspect columns, handle missing values and duplicates, and encode categorical variables.
- Engineer features by converting grades/outcomes into common risk categories and selecting academic indicators.
- Train baseline models, then compare improved ensemble models such as Random Forest, Gradient Boosting, and AdaBoost.
- Generate explanation and recommendation using feature importance or SHAP where possible.

Implementation Steps

Step	Description	Output
1	Import Dataset A and Dataset B from CSV files.	Raw data loaded.
2	Clean missing values, duplicated rows, and inconsistent formats.	Clean datasets.
3	Encode categorical variables and scale numerical variables when required.	Preprocessed features.
4	Create risk labels and align common target categories.	At Risk, In Progress, Successful.
5	Train and compare baseline and improved models.	Performance results.
6	Generate explanation and recommendation through the AI-agent layer.	Advisor-friendly output.

Evaluation Process

Metric	Purpose	Why it matters
Accuracy	Overall percentage of correct predictions.	Useful as a general indicator, but not used alone.
Precision	Correctness of predicted risk cases.	Avoids too many false warnings.
Recall	How many real at-risk students are detected.	Important for early intervention.
F1-score	Balances precision and recall.	Useful with imbalanced classes.
Confusion matrix	Shows false positives and false negatives.	Helps explain errors.
ROC-AUC / calibration	Evaluates probability quality when available.	Supports reliable risk severity levels.



The confusion matrix helps evaluate correct and missed academic-risk predictions.

6.3 Validation Strategy

- Validate Dataset A using train/test split and cross-validation if time allows.
- Train or adapt a separate model on Dataset B if direct feature matching is not possible.
- Use common risk mapping to compare outcomes across datasets.
- Compare feature importance across datasets to identify shared academic-risk factors.
- Report limitations clearly when dataset structures prevent direct transfer testing.
- Prioritize recall for at-risk detection because missing an at-risk student is more serious than flagging a student for review.

6.4 Test Plan

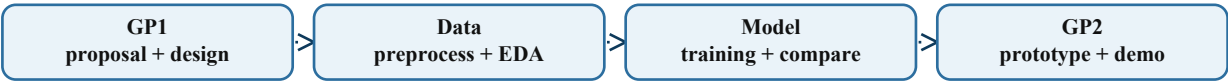
Test Case	Input	Expected Output
Data loading test	CSV dataset file	Dataset loads without errors.
Preprocessing test	Raw row with categorical and numerical values	Encoded and scaled feature vector.
Prediction test	Valid student record	Prediction class and probability.
Agent logic test	High-risk prediction	High-risk explanation and recommendation.
Interface test	User enters values	Result appears clearly on screen.
Database test	Prediction result	Record is saved or exported successfully.

CHAPTER 7: PROJECT MANAGEMENT PLAN

7.1 Team Roles and Responsibilities

Team Member	Main Responsibility
Student 1	Dataset collection, data understanding, and preprocessing.
Student 2	EDA, visualization, and dataset comparison.
Student 3	Model training, evaluation, and tuning.
Student 4	AI-agent logic, recommendations, and interface integration.
All members	Report writing, presentation, testing, and demo preparation.

Figure 7.1: Project timeline visual



The project path moves from GP1 planning to GP2 prototype and demo.

7.2 Timeline / Work Plan

GP1 Week	Milestone	Deliverable
1	Understand requirements and confirm topic	Approved project idea.
2	Collect datasets and references	Dataset links and initial reference list.
3	Problem definition and scope	Problem statement, objectives, and scope.
4	Initial proposal checkpoint	Background, motivation, baseline requirements, expected outcomes, tentative plan.
5	Data understanding and preprocessing	Data dictionary and cleaning plan.
6	EDA and initial visuals	Charts and early findings.
7	Related work and alternative approaches	Literature review and justification.
8	Midterm checkpoint	Detailed requirements, alternative solutions, selected approach, realistic plan.
9	Initial model and design	Baseline model and architecture.
10	Report and presentation preparation	Draft report and slides.
11	Final GP1 evaluation	Report, presentation, and early demo if available.
GP2 Week	Milestone	Deliverable
1	Adapt project based on GP1 feedback	Updated plan.
2	Improve preprocessing and feature engineering	Final preprocessing pipeline.
3	Train improved models	Model comparison results.
4	Validate/adapt on Dataset B	Validation and dataset comparison.
5	Add explainability and risk thresholds	Feature importance/SHAP and risk logic.
6	Implement AI agent	Recommendation module.
7	Build interface and database	Working prototype.
8	Midterm implementation evaluation	Partial implementation and preliminary results.
9	Testing and improvements	Test logs and bug fixes.
GP2 Week	Milestone	Deliverable
10	Final report and demo preparation	Final report, code, and presentation.
11	Final submission checkpoint	Complete deliverables.

7.3 Risks and Contingencies

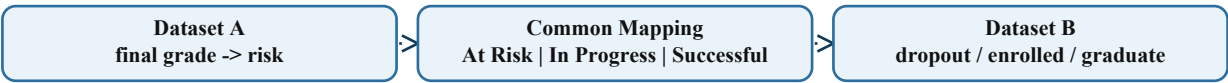
Risk	Impact	Mitigation
Dataset features do not match	Direct transfer testing may be weak.	Use target mapping, separate models, and compare outcomes.
Class imbalance in Dataset B	Model may miss dropout cases.	Use class_weight, SMOTE, or SMOTETomek.
Overfitting	Model performs poorly on unseen data.	Use train/test split, cross-validation, and simple models as baselines.
Complex model is hard to explain	Advisor may not trust predictions.	Use feature importance or SHAP and simple agent explanations.
Time limitation	Advanced features may not be fully implemented.	Prioritize baseline model, interface, and required dataset comparison first.

CHAPTER 8: DATASET COMPARISON AND EVALUATION PLAN

8.1 Dataset Differences

Comparison Point	Dataset A: UCI Student Performance	Dataset B: Student Dropout and Academic Success
Main purpose	Predict student final performance.	Predict dropout, enrolled, or graduate status.
Main target	Final grade or derived performance category.	Dropout, Enrolled, Graduate.
Context	School-level student performance.	Higher-education academic success.
Feature types	Demographic, school, social, and grade-related.	Enrollment, academic progress, financial, and demographic.
Project use	Training and initial model development.	Validation, adaptation, and comparison.
Main challenge	Grade-based target.	Outcome-based target and different feature set.

Figure 8.1: Dataset comparison visual



Dataset A and Dataset B are aligned through common outcome categories.

8.2 Common Target Mapping

Dataset	Original Target	Common Category
Dataset A	Low final grade or fail category	At Risk
Dataset A	Medium/high final grade or pass category	Successful
Dataset B	Dropout	At Risk
Dataset B	Enrolled	In Progress
Dataset B	Graduate	Successful

8.3 Initial Model Performance Comparison

Model	Dataset	Accuracy	Precision	Recall	F1-score	ROC-AUC	Notes
Logistic Regression	A	0.881	0.681	0.860	0.760	0.966	Baseline.
Decision Tree	A	0.870	0.702	0.702	0.702	0.830	Interpretable.
XGBoost / GB	A	0.916	0.843	0.754	0.796	0.971	Best A result.
Random Forest	A	0.897	0.800	0.702	0.748	0.967	Comparison.
Logistic Regression	B	0.886	0.802	0.856	0.828	0.938	Best B recall.

8.4 Required Discussion Points

- Differences between Dataset A and Dataset B in features, target variables, and educational context.
- Model performance comparison within Dataset A and Dataset B.
- Challenges of applying a model trained on one dataset to another dataset with different columns.
- Adjustments made, such as target mapping, feature selection, class balancing, or separate training.
- Interpretation of most important features and how they support academic recommendations.
- Limitations of public datasets compared with real institutional student records.

CHAPTER 9: CONCLUSION AND FUTURE WORK

This graduation project proposes a complete AI-based Academic Decision Support System that uses machine learning to predict student academic risk and support decisions. The project is strengthened by using two required datasets, following a clear data science lifecycle, applying baseline and advanced models, adding explainability, and connecting predictions to practical academic recommendations. The proposed system is suitable for GP1 because it clearly defines the problem, datasets, requirements, design, methodology, expected outcomes, and work plan. It is also suitable for GP2 because it includes a clear implementation path for the final model, AI agent, integration, interface, evaluation, and demo.

9.1 Future Work

- Add real institutional data if available and ethically approved.
- Use longitudinal student records to predict risk earlier across semesters.
- Improve the AI agent with more personalized intervention plans.
- Add advisor feedback to improve recommendations over time.

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APPENDIX A: COMPLIANCE CHECKLIST

Requirement	Covered in Report
Background and motivation	Chapter 1
Problem statement	Chapter 1
Project scope	Chapter 1
Project baseline requirements	Chapter 2
Expected outcomes	Chapter 1
Related work	Chapter 3
Alternative solutions and justification	Chapter 4
Tools and techniques	Chapter 6
Appropriate analysis	Chapters 2, 4, and 8
Design conforming to problem statement	Chapter 5
Implementation plan	Chapter 6
Validation and comparison plan	Chapters 6 and 8
Work plan	Chapter 7
Two datasets used	Chapters 2 and 8
AI agent logic	Chapter 5
Interface and database/function integration	Chapters 5 and 6
Initial model and EDA evidence	Appendix E

Appendix A Summary

This checklist confirms that the report covers the required background, problem definition, requirements, methodology, design, implementation plan, validation plan, dataset comparison, AI-agent logic, and initial model evidence.

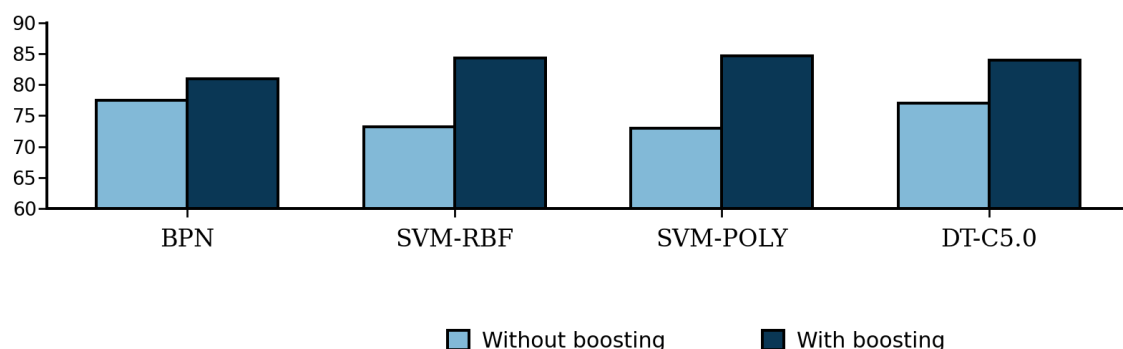
APPENDIX B: GP1 AND GP2 DELIVERABLES

Semester	Deliverable	Project Content
GP1	Problem and dataset understanding	Problem statement, scope, objectives, dataset descriptions, and EDA plan.
GP1	Data preprocessing	Cleaning, encoding, scaling, and target mapping plan.
GP1	Initial model	Baseline model and initial evaluation plan.
GP1	System design	Architecture, requirements, AI-agent logic, and interface design.
GP2	Final model	Improved and validated model.
GP2	AI agent implementation	Risk-level explanations and recommendations.
GP2	System integration	Model, function/database, and interface connected.
GP2	Dataset comparison and evaluation	Performance comparison and discussion of challenges.
GP2	Final report and demo	Final report, slides, code, and working demo.

APPENDIX C: RESEARCH-BASED FIGURES

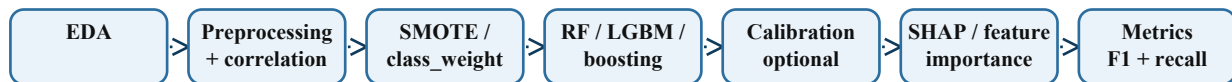
The following figures were added to support the report with visual material directly connected to the previous studies used in the literature review. They summarize relevant methods and results; they are not stock images.

Research-based model performance: F-measure before and after boosting



Source values adapted from Vafeiadis et al. (2015). Values are shown to justify comparing boosting models in this project.

Figure C.2: Research-based methodology adapted for this project

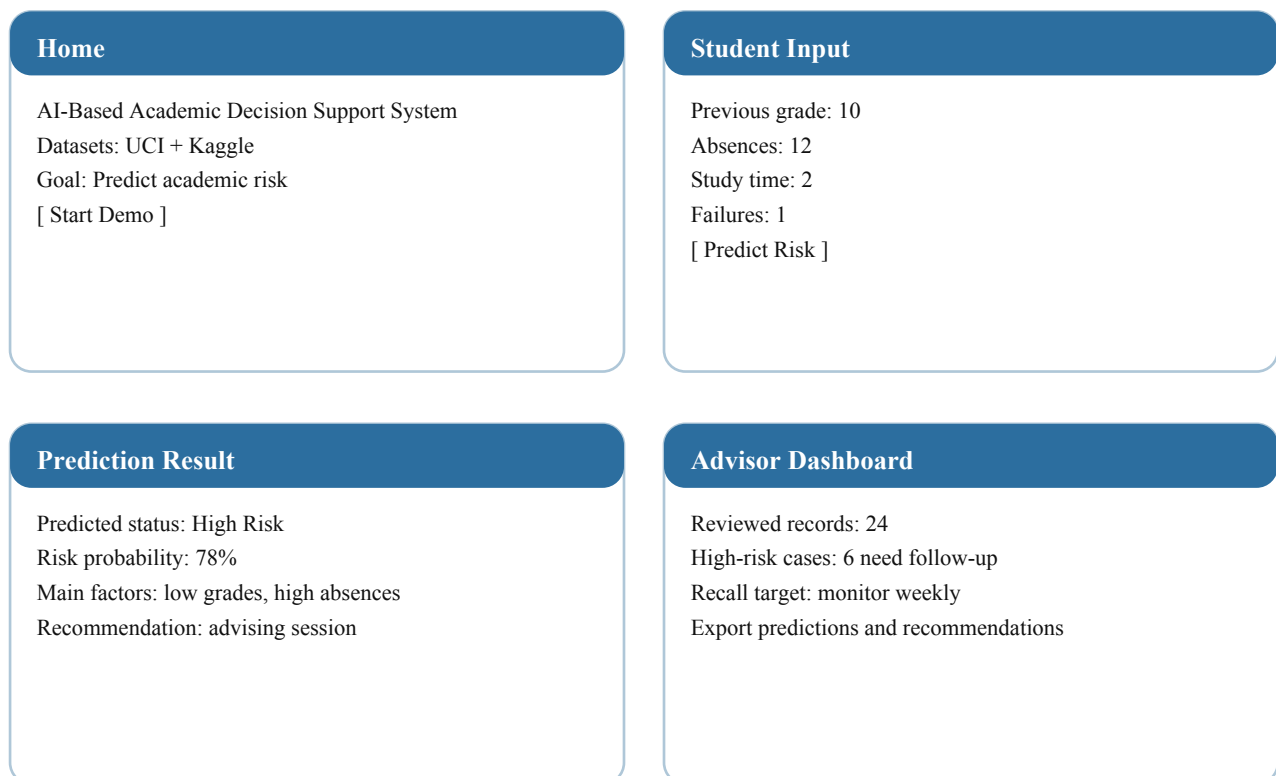


The flow reflects the data science lifecycle and modern churn-detection methodology adapted to academic risk prediction.

APPENDIX D: GP2 APPLICATION PROTOTYPE

The following mockups show how the application can look in GP2. The interface is planned as a simple academic decision-support dashboard implemented with Streamlit. These screens are planned mockups for GP2 and are not final implemented screenshots.

Planned GP2 application mockups - not final implemented screenshots



The mockups show the planned Streamlit interface for GP2 and are included to clarify the implementation direction.

Screen	Displayed Content
Home	AI-Based Academic Decision Support System; Datasets: UCI + Kaggle; Goal: Predict academic risk; Start Demo
Student Input	Previous grade; Absences; Study time; Failures; Predict Risk
Prediction Result	Predicted Status: High Risk; Risk probability: 78%; Main factors: Low grades, high absences; Recommendation
Advisor Dashboard	Reviewed: 24 records; High-risk cases: 6 need follow-up; Recall target; Export predictions and review recommendations.

APPENDIX E: IMPLEMENTATION EVIDENCE AND INITIAL MODEL RESULTS

This appendix adds GP1 implementation evidence based on the two required datasets. The results are initial prototype results, not final GP2 results. They show that the datasets were loaded, inspected, preprocessed, split into train/test sets, and evaluated using baseline and advanced models.

E.1 Dataset Loading and Summary

Dataset	Source file(s)	Rows	Columns before target mapping	Target used	Missing values	At-risk records	At-risk rate
Dataset A - UCI Student Performance	student-mat.csv + student-por.csv	1044	33	G3 mapped to academic_risk (G3 < 10 = At Risk)	0	230	22.03
Dataset B - Student Dropout and Academic Success	dataset.csv	4424	35	Target mapped to academic_risk (Dropout = At Risk)	0	1421	32.12

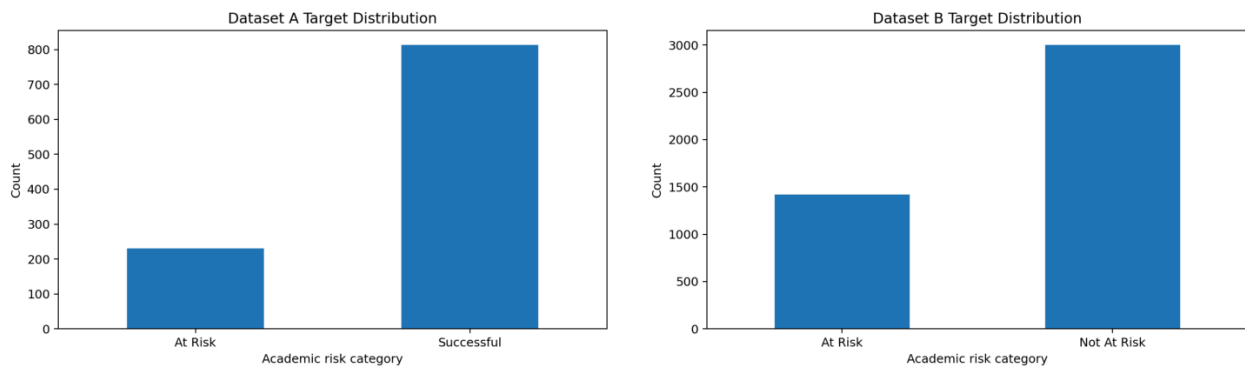
Dataset A combines student-mat.csv and student-por.csv from the UCI Student Performance dataset. Dataset A uses final grade G3 to create a binary academic-risk target: G3 < 10 is treated as At Risk. Dataset B uses the original Target field, where Dropout is mapped as At Risk for initial binary evaluation.

E.2 Initial EDA Evidence

Evidence	Purpose
Dataset A target distribution	Shows class balance after mapping G3 to risk categories.
Dataset B target distribution	Shows dropout-risk class balance after mapping Target to risk categories.
Dataset A final grade distribution	Shows how final grades are distributed before target mapping.
Missing-values checks	Confirms whether imputation is needed before model training.

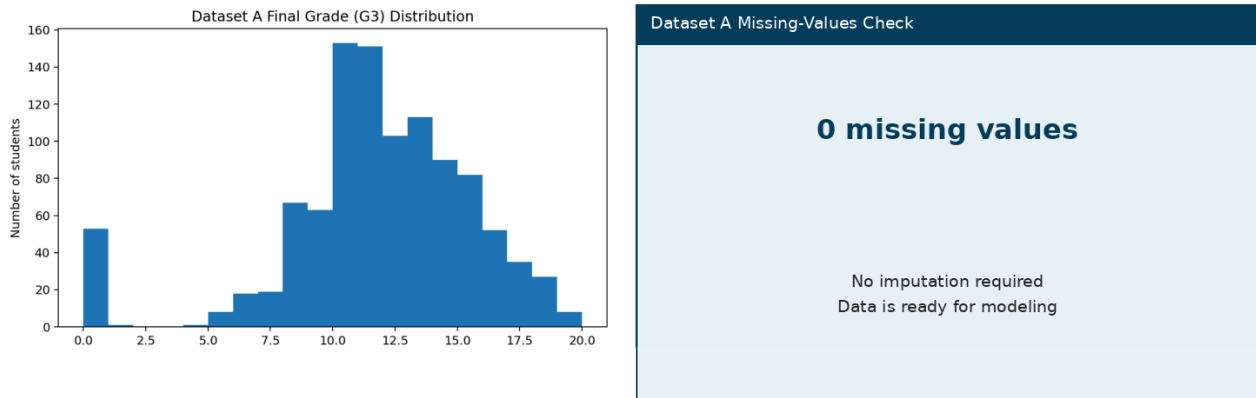
E.3 EDA Visual Evidence

Figure E.1: Dataset A and Dataset B Target Distributions



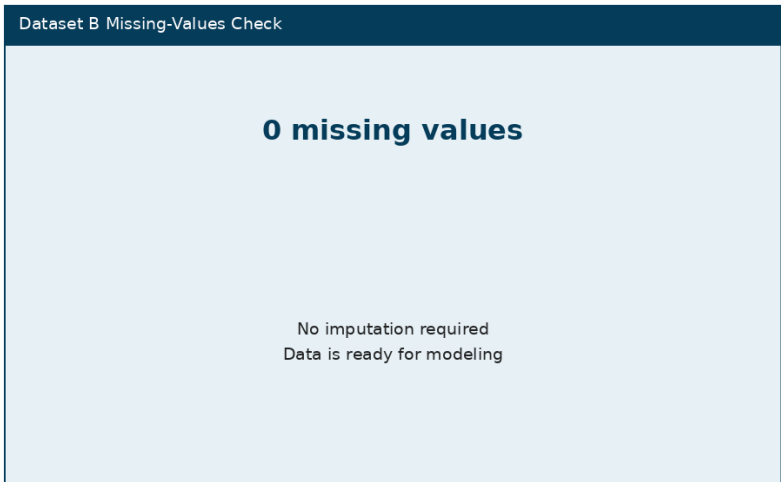
Both datasets were converted into academic-risk categories for initial binary classification. Dataset B has a higher at-risk proportion, which is important when selecting recall-focused metrics.

Figure E.2: Dataset A Final Grade Distribution and Missing-Values Check



The G3 grade distribution supports the risk-threshold mapping used for Dataset A. The missing-values check shows that no missing-value imputation was required for the loaded UCI files.

Figure E.3: Dataset B Missing-Values Check



The Dataset B file used in this run did not require missing-value imputation before the initial modeling pipeline.

E.4 Preprocessing and Modeling Setup

The implementation uses a supervised-learning workflow: target mapping, train/test split with stratification, categorical encoding through one-hot encoding, scaling for Logistic Regression, and tree-based modeling for Decision Tree, Random Forest, and XGBoost. Recall and F1-score are prioritized because missing an at-risk student is more serious than flagging a student for advisor review.

Step	Implementation Evidence
Target mapping	Dataset A: $G3 < 10$ = At Risk; Dataset B: Dropout = At Risk
Feature preparation	One-hot encoding for categorical features; numeric features passed through or scaled depending on model
Data split	75% training and 25% testing with stratified class distribution
Baseline models	Logistic Regression and Decision Tree
Advanced/comparison models	Random Forest and XGBoost Classifier
Evaluation metrics	Accuracy, Precision, Recall, F1-score, ROC-AUC, and confusion matrix

E.5 Initial Model Results

Dataset	Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	Test rows
Dataset A	Logistic Regression	0.881	0.681	0.860	0.760	0.966	261
Dataset A	Decision Tree	0.870	0.702	0.702	0.702	0.830	261
Dataset A	Random Forest	0.897	0.800	0.702	0.748	0.967	261
Dataset A	XGBoost Classifier	0.916	0.843	0.754	0.796	0.971	261
Dataset B	Logistic Regression	0.886	0.802	0.856	0.828	0.938	1106
Dataset B	Decision Tree	0.841	0.734	0.792	0.762	0.886	1106
Dataset B	Random Forest	0.887	0.891	0.738	0.807	0.937	1106
Dataset B	XGBoost Classifier	0.883	0.867	0.752	0.805	0.937	1106

Interpretation: On Dataset A, XGBoost produced the highest initial F1-score and ROC-AUC among the tested models. On Dataset B, Logistic Regression achieved the strongest initial Recall, while Random Forest and XGBoost remained competitive. These numbers are suitable as GP1 evidence and should be refined in GP2 using tuning, cross-validation, and final validation.

E.6 Confusion Matrix Evidence

The following confusion matrices show how the selected initial models performed on the held-out test sets. They help identify false negatives, which are especially important in early-warning academic decision support.

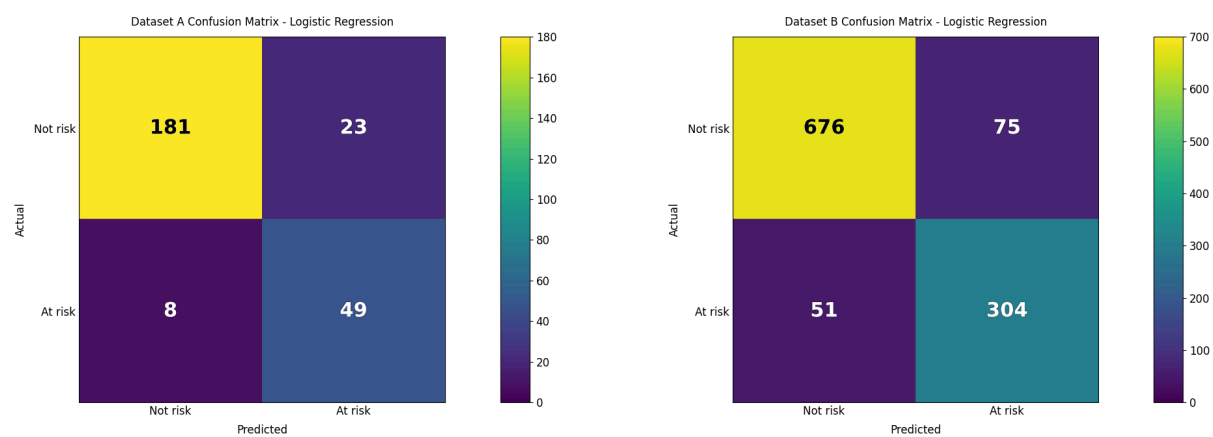


Figure E.4: Confusion matrices for the selected initial models on each dataset.

E.7 Feature Importance Evidence

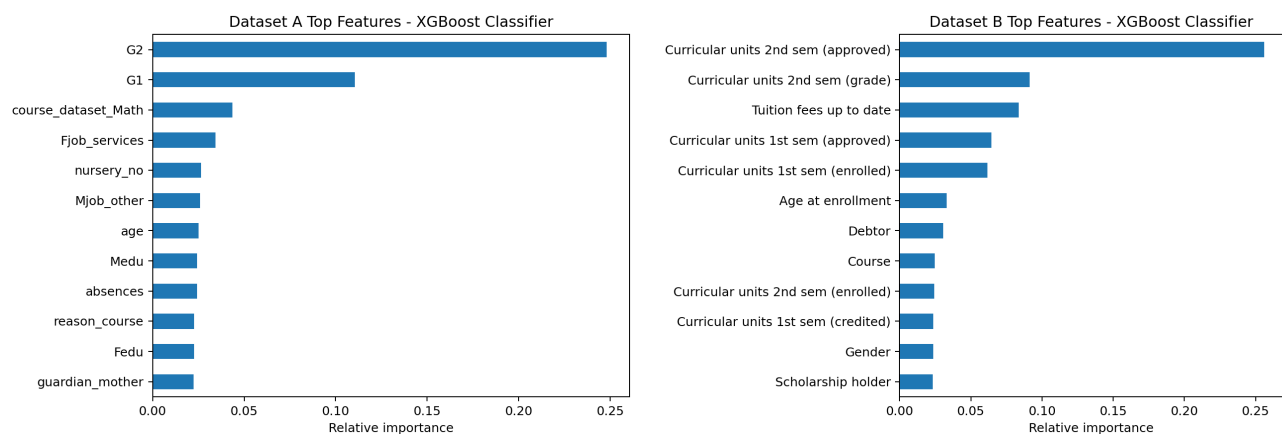
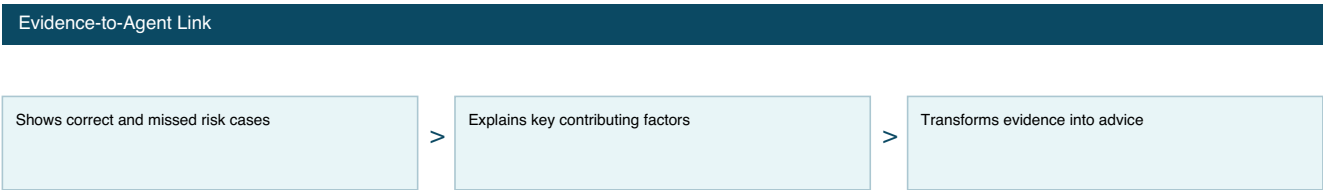


Figure E.5: Feature-importance charts generated from the XGBoost model. These factors can support the AI-agent explanation layer.



E.8 Top Factors for Agent Explanation

The AI agent should not only return a class label. It should translate important features into a simple explanation and recommendation. The following tables summarize the strongest initial feature signals that can support advisor-friendly explanations.

Dataset A - Top Features

Feature	Importance
G2	0.248
G1	0.110
course_dataset_Math	0.044
Fjob_services	0.034
nursery_no	0.026
Mjob_other	0.026
age	0.025
Medu	0.024

Dataset B - Top Features

Feature	Importance
Curricular units 2nd sem (approved)	0.256
Curricular units 2nd sem (grade)	0.091
Tuition fees up to date	0.084
Curricular units 1st sem (approved)	0.064
Curricular units 1st sem (enrolled)	0.062
Age at enrollment	0.033
Debtor	0.031
Course	0.025

E.9 GP1 Evidence Conclusion

This evidence confirms that the project has moved beyond planning: both required datasets were loaded, checked, mapped into academic-risk categories, preprocessed, and evaluated with multiple initial models. The results justify keeping XGBoost as an advanced comparison model while retaining interpretable baselines. In GP2, the team should improve these results through parameter tuning, cross-validation, interface integration, database saving, and AI-agent recommendation testing.

Final Evidence Conclusion

Point	Conclusion
Model evidence	The report now includes actual model outputs for both datasets.
Project readiness	The GP1 report demonstrates planning, implementation, and validation evidence.
Remaining work	GP2 should focus on prototype integration, testing, and advisor-facing interface.